

Research Review and Revision Guide

Curriculum Learning $\times$ MaxRL

Mathematical proofs, reviewer questions, and rewriting guidance

Internal research review workbook

Updated 2026-08-04

Evidence base. This guide reviews the 17-page working draft *The Estimator Decides: What Curricula and Failure Recycling Can and Cannot Do in RL with Verifiable Rewards* (working draft v0.9, August 4, 2026) against Tajwar, Zeng, et al., *Maximum Likelihood Reinforcement Learning* (arXiv:2602.02710v1). It also consolidates the earlier Markdown review into a self-contained mathematical and writing document.

Scope. This is a paper review and derivation audit. It is not an independent reproduction of the repository, a formal verification of every implementation path, or an exhaustive novelty search over all 2025–2026 literature. Whenever the source files do not establish a claim, the guide labels the point as a reviewer inference, a proposed theorem, or a requested experiment rather than silently treating it as fact.

Main takeaway

There is a strong paper here. Its most defensible thesis is narrower and deeper than the current broad slogan: *the practical finite-group estimator defines an exact activity profile that constrains where prompt allocation can produce reward contrast; task transfer remains structure-dependent; relabeling can create contrast beyond allocation, but its destination distribution can also concentrate the policy and reduce multi-sample coverage.*

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Part I

Executive review and revision priorities

1 Overall assessment

The draft contains a genuine conceptual contribution. MaxRL derives how a task is weighted in expectation after it is sampled. Your extension asks an earlier, data-level question: before generation compute is spent, how likely is a finite rollout group to produce a contrastive update under the estimator that will consume it? For the practical centered MaxRL implementation, that question has a closed-form answer. This connects four levels that are usually discussed separately:

1. the population objective and its pass-rate weight function;
2. the finite-group event structure, especially all-fail and all-success groups;
3. the curriculum’s pre-rollout allocation decision;
4. the destination distribution created by hindsight relabeling.

The project also demonstrates unusually good scientific discipline. The draft reports nulls, identifies a posterior-feedback failure, distinguishes wall-clock from step-matched effects, discloses a gate decay bug, retracts an oracle comparison, and includes a suite that defeats the broad safety story. Those are strengths. The remaining task is to make the claims match the exact estimator, sampling law, statistical unit, and evidence level.

1.1 Central risks in the current version

Priority

The highest-risk statements are not the equations themselves; they are the nouns attached to them. The draft repeatedly calls coefficient mass “the expected learning signal,” calls an implementation-dependent all-fail convention a universal dead zone, calls a coverage main effect a curriculum-by-estimator interaction, and calls semantically valid relabels exact destination ML gradients. Each can be repaired without abandoning the core idea.

The main revision risks are:

1. **Quantity naming.** The exact theorem computes coefficient or advantage mass, not gradient norm, signal-to-noise ratio, or expected improvement. The right repair is an exact factorization theorem, not only weaker prose.
2. **Estimator ambiguity.** The MaxRL baseline contains a raw estimator, a full control-variate estimator, and a practical algorithm that drops the all-fail control-variate term. They do not have the same objective order or coefficient-mass profile.
3. **Interaction identification.** The maze supports an estimator-conditioned coverage main effect across curriculum variants. It does not, by itself, prove that adding the teacher harms GRPO.

4. **Hindsight sampling law.** Exact verification and prompt rewriting establish semantic validity, but source-generated data are generally off-policy for the destination prompt.
5. **Gate evidence.** The three-seed gate operating point was produced by a disclosed decay bug; the corrected setting currently has one seed in the draft.
6. **Statistical units.** The manuscript alternates between five and eighteen MaxRL observations and calls six configuration-seed contrasts “six seeds.” The run inventory and clustering must be explicit.
7. **Narrative breadth.** Nine experimental rungs, preliminary LLM cells, classic control, IsaacLab, inference multipliers, and several audit stories compete with the central theorem.
8. **Anonymity.** The anonymous manuscript links to a public repository whose URL reveals the author identity.

2 Revision priorities

2.1 P0: submission-blocking changes

- P0.1. Anonymize the paper and artifacts.** Replace the public repository URL, scrub usernames and metadata, and inspect the PDF, supplementary archive, generated JSON, commit history, and code comments.
- P0.2. Distinguish all three MaxRL estimators.** State explicitly that Tajwar et al.’s Theorem 2 is correct for the raw estimator, while your $T = N - 1$ result concerns the practical centered implementation that zeros the entire $K = 0$ group.
- P0.3. Fix the factor-of-two notation.** Reserve A_N for full coefficient mass and $\nu_N = A_N/2$ for the normalized sampling utility.
- P0.4. Replace “exact expected learning signal.”** Add the coefficient-mass factorization in section 8; call ν_N the estimator-controlled scalar multiplying the success-versus-failure score contrast.
- P0.5. Scope the dead-zone statement.** The full control-variate estimator emits a nonzero sample update on $K = 0$. Run or discuss that missing baseline before saying recycling is the only possible channel.
- P0.6. Rewrite the hindsight theorem.** Present it as a verified destination semi-gradient under a source-induced proposal law, exact only under a measurable law-matching condition.
- P0.7. Separate main effect from interaction.** Use the maze for the estimator main effect; reserve the stronger teacher-harms-GRPO claim for a powered, schedule-matched factorial experiment.
- P0.8. Reanalyze statistical units.** Publish a one-row-per-run registry and use seed/warmstart clusters in permutation or bootstrap inference.
- P0.9. Rerun the corrected gate.** The final coverage-restoration claim should use the intended decay and multiple paired seeds.

2.2 P1: decisive experiments

- P1.1.** Practical drop- $K = 0$ MaxRL versus full-control-variate MaxRL on a chain, the frontier-heavy suite, and one maze setting.
- P1.2.** A frozen or externally generated prompt schedule replayed identically under MaxRL and GRPO.
- P1.3.** A corrected gate-strength sweep with three to five paired seeds and destination-pass-rate telemetry.
- P1.4.** A uniform-with-replacement GSM8K control to isolate repetition from curriculum steering.
- P1.5.** Generated-token or environment-transition matching in addition to wall-clock and optimizer-step matching.
- P1.6.** A source-versus-destination distribution-shift diagnostic and, ideally, a cross-fitted relabel ablation.

2.3 P2: narrative and presentation

- P2.1.** Reduce the main paper to three decisive experiment stories plus one boundary-condition synthesis.
- P2.2.** Move preliminary GSM8K interaction evidence, the single-checkpoint $11\times$ threshold result, detailed classic-control curves, and most audit history to the appendix unless the decisive replications finish.
- P2.3.** Replace recurring slogans with one stable vocabulary: coefficient mass, estimator activity, operationally dead regime, coverage-stable, sampling premium, one-step mass oracle, and destination semi-gradient.
- P2.4.** Make every figure caption state the estimator convention, group size, statistical unit, and whether the schedule is matched or merely generated by the same rule.

3 The strongest defensible thesis

Recommended one-sentence thesis

For a fixed group estimator, its finite-sample coefficient profile determines where a sampled prompt can produce reward contrast; this profile can guide allocation, but it does not determine cross-task transfer, and hindsight relabeling can create contrast beyond allocation while shifting training toward destinations that may reduce multi-sample coverage.

A shorter hook, positioned directly against MaxRL, is:

Recommended opening contrast with MaxRL

MaxRL derives how a prompt is weighted after it is sampled. We derive the finite-group activity profile that tells a curriculum whether sampling that prompt can produce contrast at all.

4 Claim-to-evidence boundary

Current claim	Why a reviewer may object	Defensible rewrite now	Evidence needed for the stronger claim
“Expected learning signal is exactly $2(\text{pass}@N - \text{pass}@1)$.”	The theorem computes $\mathbb{E} \sum_i w_i $, not gradient norm, SNR, or improvement.	“The expected coefficient mass is exact, and its half-mass is the scalar multiplying the success–failure score contrast for balanced binary estimators.”	Direct bridge to expected improvement or a task-transfer model.
“The dead zone no sampler can reach.”	A sampled all-fail group can receive a nonzero full-control-variate update; small $p > 0$ can also be reached with enough budget.	“Under the practical zero-constant-group estimator, all-fail groups are silent; at a fixed (B, N) budget, low- p tasks can be operationally dead.”	Full-CV baseline and a budget-indexed definition.
“Recycling is the only channel into the dead zone.”	This is only true with the practical estimator held fixed and after excluding other all-fail updates.	“Relabeling is the signal-creation mechanism studied here; re-allocation alone cannot create contrast for a dropped all-fail group.”	Compare full-CV, unlikelihood, entropy, or other negative-only updates.
“The same curriculum degrades GRPO.”	The maze mainly shows an estimator coverage main effect; adaptive teachers produce different realized schedules.	“Coverage behavior separates by estimator across curriculum variants; the teacher interaction remains suggestive.”	Same realized schedule, paired factorial design, adequate seeds, delivered steering.
“Hindsight yields the exact ML gradient of the achieved task.”	Source-generated trajectories are not generally sampled from the destination-conditioned policy.	“Hindsight yields a verified destination semi-gradient under the source-induced proposal; it matches the fresh destination update under a conditional-law matching condition.”	Fresh-destination comparison, distribution-shift diagnostic, or cross-fitting.
“Perfect pass-rate information is worth almost nothing.”	Oracle-only strongly exceeds posterior-only in the reported chain; oracle ties posterior plus recycling, not posterior alone.	“Recycling can compensate for allocation error; perfect allocation still improves substantially over posterior-only allocation.”	Matched oracle-only versus posterior-only analysis.
“The gate restores coverage.”	The intended setting has one corrected seed; three-seed runs used an unintended effective threshold.	“An unintended intermediate gate strength restores frontier coverage in three seeds; a corrected one-seed run shows the stronger endpoint.”	Corrected multi-seed sweep.
“Confirmed at LLM scale.”	Two seeds, weak steering, replacement confound, and one regression-shape replication.	“The LLM experiment diagnoses posterior starvation and yields a same-sign endpoint contrast under GRPO, with magnitude tracking delivered steering.”	Strong steering, replacement control, three to five paired seeds.
“No new difficulty hyperparameters.”	Band center is derived, but decay, floor, concentration, dose, and gate strength remain.	“No hand-set target difficulty or band width is required; several implementation and safety knobs remain.”	None; this is the accurate framing.

Part II

Mathematical audit and complete derivations

5 Setup and notation

Fix a task or prompt x . Let

$$z_i \stackrel{\text{i.i.d.}}{\sim} m_\theta(\cdot \mid x), \quad r_i = R(x, z_i) \in \{0, 1\}, \quad K = \sum_{i=1}^N r_i,$$

with score

$$S_i = \nabla_\theta \log m_\theta(z_i \mid x).$$

Write

$$p = p_\theta(x) = \mathbb{P}(r = 1 \mid x), \quad q = 1 - p,$$

and

$$\text{pass@}k(p) = 1 - q^k.$$

The MaxRL truncated objective of order T has gradient

$$\nabla_\theta J_{\text{MaxRL}}^{(T)}(x) = \sum_{j=0}^{T-1} q^j \nabla_\theta p = \frac{1 - q^T}{p} \nabla_\theta p.$$

All exact score identities below assume $0 < p < 1$ and extend to endpoints by limits when the model is smooth and has full support.

6 The baseline contains three different estimators

The most important correction in the paper is conceptual, not adversarial: Tajwar et al.’s raw theorem, their full control-variate expression, and their practical algorithm are distinct estimators.

6.1 Raw success-conditioned estimator

The baseline’s Eq. (9) and Theorem 2 use

$$\hat{g}_N^{\text{raw}} = \mathbf{1}\{K \geq 1\} \frac{1}{K} \sum_{i=1}^N r_i S_i. \tag{1}$$

This estimator is unbiased for order $T = N$.

6.2 Full control-variate estimator

The baseline’s Eq. (10) subtracts the unconditional average score:

$$\hat{g}_N^{\text{full}} = \mathbf{1}\{K \geq 1\} \frac{1}{K} \sum_{i=1}^N r_i S_i - \frac{1}{N} \sum_{i=1}^N S_i. \quad (2)$$

The second term is retained at $K = 0$. Since $\mathbb{E}[S_i] = 0$, this remains unbiased for $T = N$.

6.3 Practical centered, drop-all-fail estimator

The baseline’s practical Algorithm 1 drops both terms when $K = 0$:

$$\hat{g}_N^{\text{prac}} = \mathbf{1}\{K \geq 1\} \sum_{i=1}^N \left(\frac{r_i}{K} - \frac{1}{N} \right) S_i. \quad (3)$$

This is the estimator analyzed by the working draft. It is unbiased for order $T = N - 1$, not $T = N$.

Table 2: The three MaxRL-related estimators should be named separately. Here “mass” means $\mathbb{E} \sum_i |w_i|$.

Estimator	$K = 0$ behavior	Objective order	Expected coefficient mass	Endpoint profile
Raw success average	zero	$T = N$	$1 - q^N$	zero only at $p = 0$
Full control variate	$-N^{-1} \sum_i S_i$	$T = N$	$2q - q^N$	nonzero as $p \downarrow 0$
Practical centered/drop	zero	$T = N - 1$	$2(q - q^N)$	zero at $p = 0$ and $p = 1$

Evidence boundary

The two-ended, zone-of-proximal-development-shaped coefficient profile is not a generic property of every success-conditioned estimator. It is created by the combination of centering and the practical convention that discards the entire all-fail group.

7 Conditional score identities

Lemma 7.1 (Success and failure score means). *Define*

$$\mu_+(x) = \mathbb{E}[S \mid r = 1, x], \quad \mu_-(x) = \mathbb{E}[S \mid r = 0, x].$$

Then

$$\mu_+(x) = \frac{\nabla_{\theta} p}{p}, \quad \mu_-(x) = -\frac{\nabla_{\theta} p}{q}, \quad (4)$$

and therefore

$$\mu_+(x) - \mu_-(x) = \frac{\nabla_{\theta} p}{pq}. \quad (5)$$

Proof. The likelihood-ratio identity gives

$$\nabla_{\theta} p = \nabla_{\theta} \mathbb{E}[r] = \mathbb{E}[rS] = p \mathbb{E}[S \mid r = 1] = p\mu_+.$$

Also $\mathbb{E}[S] = 0$, so

$$0 = p\mu_+ + q\mu_-.$$

Substituting $p\mu_+ = \nabla_{\theta} p$ yields $q\mu_- = -\nabla_{\theta} p$. Subtracting the two conditional means gives eq. (5). $\square$

8 A general coefficient-mass factorization

The strongest repair to the phrase “expected learning signal” is an exact theorem for the family of balanced binary group estimators that includes practical MaxRL, RLOO, and finite-group GRPO.

Definition 8.1 (Balanced binary group estimator). For $1 \leq K = k \leq N - 1$, suppose a symmetric group estimator assigns each success coefficient a_k/k and each failure coefficient $-a_k/(N - k)$, where $a_k \geq 0$, and assigns zero coefficients to constant groups $K \in \{0, N\}$. Then its full coefficient mass conditional on $K = k$ is $2a_k$.

Theorem 8.2 (Coefficient-mass factorization). *For a balanced binary group estimator,*

$$\mathbb{E}[\hat{g} \mid x] = \nu_N(p)(\mu_+(x) - \mu_-(x)), \quad \nu_N(p) := \mathbb{E}[a_K] = \frac{1}{2}\mathbb{E}\left[\sum_i |w_i|\right]. \quad (6)$$

Equivalently,

$$\mathbb{E}[\hat{g} \mid x] = \frac{\nu_N(p)}{p(1-p)} \nabla_{\theta} p. \quad (7)$$

Proof. Condition on $K = k$ with $1 \leq k \leq N - 1$. The total positive coefficient is

$$k \frac{a_k}{k} = a_k,$$

and the total negative coefficient is

$$(N - k) \left(-\frac{a_k}{N - k} \right) = -a_k.$$

By exchangeability, successful trajectories have conditional score mean μ_+ and failed trajectories have conditional score mean μ_- . Therefore

$$\mathbb{E}[\hat{g} \mid K = k, x] = a_k\mu_+ - a_k\mu_- = a_k(\mu_+ - \mu_-).$$

Constant groups contribute zero. Averaging over K yields eq. (6); substituting eq. (5) yields eq. (7). $\square$

Main takeaway

The exact statement is not that coefficient mass equals the whole learning signal. It is that normalized half-mass $\nu_N(p)$ is the estimator-controlled scalar multiplying the task-specific success–failure score separation. Score geometry and cross-task transfer still matter, but ν_N is more than an arbitrary heuristic.

9 Exact results for the three MaxRL variants

9.1 Raw estimator: order $T = N$

Proposition 9.1 (Raw estimator expectation). *For eq. (1),*

$$\mathbb{E}[\hat{g}_N^{\text{raw}} \mid x] = (1 - q^N) \mu_+ = \frac{1 - q^N}{p} \nabla_{\theta} p = \sum_{j=0}^{N-1} q^j \nabla_{\theta} p. \quad (8)$$

Thus the raw estimator is unbiased for $J_{\text{MaxRL}}^{(N)}$.

Proof. Conditional on $K \geq 1$, the average score over successful samples has expectation μ_+ . The event occurs with probability $1 - q^N$. Substitute $\mu_+ = \nabla_{\theta} p / p$ and expand the geometric series. $\square$

Its expected coefficient mass is simply

$$\mathbb{E} \sum_i |w_i| = \mathbb{P}(K \geq 1) = 1 - q^N.$$

This mass does not vanish at $p = 1$, illustrating why mass alone cannot be called the entire gradient: at $p = 1$, the conditional success law equals the unconditional law and its mean score is zero even though the raw coefficients sum to one.

9.2 Full control variate: order $T = N$ and a nonzero all-fail update

Proposition 9.2 (Full control-variate mass). *For eq. (2),*

$$\mathbb{E}[\hat{g}_N^{\text{full}} \mid x] = \mathbb{E}[\hat{g}_N^{\text{raw}} \mid x], \quad (9)$$

and

$$\mathbb{E} \left[\sum_i |w_i| \right] = 2q - q^N. \quad (10)$$

Moreover,

$$\mathbb{E}[\hat{g}_N^{\text{full}} \mid K = 0, x] = -\mu_-(x) = \frac{\nabla_{\theta} p}{q}. \quad (11)$$

Proof. The unconditional average score has mean zero, so subtracting it preserves the raw estimator's expectation. When $K \geq 1$, the coefficient mass is $2(1 - K/N)$. When $K = 0$, all N coefficients equal $-1/N$, so the mass is one. Therefore

$$\begin{aligned} \mathbb{E} \sum_i |w_i| &= q^N + 2\mathbb{E} \left[\left(1 - \frac{K}{N}\right) \mathbf{1}\{K \geq 1\} \right] \\ &= q^N + 2(1 - q^N - p) \\ &= 2q - q^N. \end{aligned}$$

Conditional on $K = 0$, each score has mean μ_- ; hence the negative average score has mean $-\mu_-$. Apply eq. (4). $\square$

Priority

This estimator is the cleanest mathematical challenge to an unconditional statement that “no loss-level update can act on an all-fail group.” The practical algorithm may still outperform it because the conditional update can be noisy or unstable. That is an empirical question and a valuable baseline, not a reason to omit the distinction.

9.3 Practical centered estimator: order $T = N - 1$

For $1 \leq K = k \leq N - 1$, the total positive and negative coefficient magnitudes of eq. (3) are both

$$a_k = 1 - \frac{k}{N}.$$

Therefore it is balanced and theorem 8.2 applies.

Theorem 9.3 (Practical MaxRL activity and objective order). *For $N \geq 2$, define*

$$A_N(p) := \mathbb{E} \left[\sum_i |w_i| \right], \quad \nu_N(p) := \frac{1}{2} A_N(p). \quad (12)$$

Then the practical centered, drop-all-fail estimator satisfies

$$\nu_N(p) = \mathbb{P}(K \geq 1) - \frac{\mathbb{E}K}{N} = q - q^N = \text{pass@}N(p) - \text{pass@}1(p), \quad (13)$$

$$A_N(p) = 2(q - q^N), \quad (14)$$

$$\mathbb{E}[\hat{g}_N^{\text{prac}} \mid x] = \nu_N(p)(\mu_+ - \mu_-) = \frac{1 - q^{N-1}}{p} \nabla_{\theta} p = \sum_{j=0}^{N-2} q^j \nabla_{\theta} p. \quad (15)$$

Consequently, the practical estimator is unbiased for $J_{\text{MaxRL}}^{(N-1)}$.

Proof. The normalized half-mass is

$$\begin{aligned} \mathbb{E}[a_K] &= \mathbb{E} \left[\left(1 - \frac{K}{N} \right) \mathbf{1}\{K \geq 1\} \right] \\ &= \mathbb{P}(K \geq 1) - \frac{\mathbb{E}K}{N} \\ &= (1 - q^N) - p \\ &= q - q^N. \end{aligned}$$

The full mass is twice this value. The factorization theorem gives

$$\mathbb{E}[\hat{g}_N^{\text{prac}} \mid x] = (q - q^N) \frac{\nabla_{\theta} p}{pq} = \frac{1 - q^{N-1}}{p} \nabla_{\theta} p.$$

The final equality follows from the geometric series

$$\frac{1 - q^{N-1}}{1 - q} = \sum_{j=0}^{N-2} q^j.$$

Since $p = 1 - q$, the estimator targets the order- $N - 1$ truncated objective. $\square$

Recommended theorem interpretation

The quantity $\nu_N(p) = \text{pass}@N(p) - \text{pass}@1(p)$ is exactly the estimator-controlled scalar multiplying the success–failure score contrast. It is the probability that a task is solved within N attempts but not in one, but it is not by itself the gradient norm or the expected improvement of training on that task.

9.4 Why the $T = N - 1$ refinement is collegial, not a correction of Theorem 2

The precise sentence to use is:

Ready-to-use rewrite

Tajwar et al.’s Theorem 2 is exact for the raw success-conditioned estimator in their Eq. (9). Their practical Algorithm 1 additionally zeroes the all-fail control-variate term; that implementation corresponds to truncation order $T = N - 1$.

Avoid phrases such as “the MaxRL theorem is wrong.” The contribution is an implementation-level refinement that matters for your finite-group activity accounting.

10 Shape, peak, and operational regimes

Proposition 10.1 (Strict concavity and peak). *For $N \geq 2$,*

$$\nu_N(p) = q - q^N$$

is strictly concave on $(0, 1)$, vanishes at $p = 0$ and $p = 1$, and has the unique maximizer

$$p_N^* = 1 - N^{-1/(N-1)}. \quad (16)$$

At the maximum,

$$\nu_N(p_N^*) = \left(1 - \frac{1}{N}\right) N^{-1/(N-1)}. \quad (17)$$

Proof. Differentiate with $q = 1 - p$:

$$\nu_N'(p) = -1 + Nq^{N-1}, \quad \nu_N''(p) = -N(N-1)q^{N-2} < 0.$$

Thus there is one critical point and it is the strict maximum. Solving $Nq^{N-1} = 1$ gives $q^* = N^{-1/(N-1)}$ and eq. (16). Since $(q^*)^N = q^*/N$, the peak value is $q^*(1 - 1/N)$. $\square$

For large N , writing $a_N = \log N/(N-1)$,

$$p_N^* = 1 - e^{-a_N} = \frac{\log N}{N-1} - \frac{(\log N)^2}{2(N-1)^2} + O\left(\frac{(\log N)^3}{N^3}\right). \quad (18)$$

The common shorthand $p_N^* \approx \log N/N$ is acceptable in prose, but the theorem and figure caption should retain the exact expression.

Table 3: Exact activity peak for common rollout counts.

N	p_N^*	$\nu_N(p_N^*)$
2	0.5000	0.2500
4	0.3700	0.4725
8	0.2570	0.6501
16	0.1688	0.7793
32	0.1058	0.8663
64	0.0639	0.9215

10.1 Tail expansions

As $p \downarrow 0$,

$$\nu_N(p) = (N-1)p - \binom{N}{2}p^2 + O(p^3). \quad (19)$$

As $p \uparrow 1$, with $q = 1 - p$,

$$\nu_N(p) = q - q^N \sim q. \quad (20)$$

Thus the hard-tail activity grows by a factor of approximately $N - 1$ relative to the $N = 2$ slice, while the mastered tail decays linearly.

10.2 Do not call two point zeros a uniquely defined band width

The endpoint zeros and unique interior maximum induce three useful operational regimes, but a mathematical “band width” requires a level-set convention. One clean definition is

$$\mathcal{B}_{N,\eta} = \{p \in [0, 1] : \nu_N(p) \geq \eta \nu_N(p_N^*)\}, \quad 0 < \eta < 1. \quad (21)$$

Because η is a choice, the strongest hyperparameter-free statement is:

Ready-to-use rewrite

The rollout count N determines the utility profile and its peak without requiring a hand-set target difficulty or band width.

11 A budget-indexed definition of an operationally dead task

For the practical centered estimator, a group has nonzero coefficients exactly when $0 < K < N$. Define the live-group probability

$$\ell_N(p) = \mathbb{P}(0 < K < N) = 1 - p^N - q^N. \quad (22)$$

If a task receives B independent groups, the probability that it produces no live group is

$$\mathbb{P}(\text{no live group in } B \text{ attempts}) = (1 - \ell_N(p))^B. \quad (23)$$

Definition 11.1 (Operationally δ -dead at budget (B, N)). A task is operationally δ -dead if

$$1 - (1 - \ell_N(p))^B \leq \delta, \quad (24)$$

meaning that the probability of observing even one live contrastive group within the allocated budget is at most δ .

For the hard tail, $\ell_N(p) = Np + O(p^2)$; for the mastered tail, $\ell_N(p) = Nq + O(q^2)$. The approximation $B\ell_N(p) \ll 1$ gives a simple budget diagnostic.

Evidence boundary

At $p = 0$ or $p = 1$, the practical estimator is structurally silent. At small but nonzero p , the task is not mathematically unreachable; it may be operationally dead at the deployed rollout and training budget. Use “operationally dead” rather than treating every low-pass-rate task as a literal zero.

12 RLOO and finite-group GRPO

12.1 RLOO is the $N = 2$ activity slice

Use the convention

$$\hat{g}_N^{\text{RLOO}} = \frac{1}{N} \sum_{i=1}^N \left(r_i - \frac{1}{N-1} \sum_{j \neq i} r_j \right) S_i. \quad (25)$$

For a group with $K = k$, the aggregate positive coefficient is

$$a_k^{\text{RLOO}} = \frac{k(N-k)}{N(N-1)}. \quad (26)$$

Therefore

$$\nu_N^{\text{RLOO}}(p) = \mathbb{E}[a_K^{\text{RLOO}}] = pq, \quad \mathbb{E} \sum_i |w_i| = 2pq. \quad (27)$$

The factorization theorem gives

$$\mathbb{E}[\hat{g}_N^{\text{RLOO}} \mid x] = \nabla_{\theta} p,$$

as required for expected-reward RL.

For practical MaxRL at $N = 2$,

$$\nu_2(p) = q - q^2 = pq,$$

so the learnability profile $p(1-p)$ is exactly the $N = 2$ practical-MaxRL activity profile under this normalization.

12.2 Finite-group GRPO

Assume GRPO uses the population standard deviation within the group and averages the trajectory updates by $1/N$:

$$\hat{g}_N^{\text{GRPO}} = \frac{1}{N} \sum_{i=1}^N \frac{r_i - K/N}{\sqrt{(K/N)(1-K/N)}} S_i, \quad 0 < K < N, \quad (28)$$

with zero update on constant groups. For $K = k$, the aggregate positive coefficient is

$$a_k^{\text{GRPO}} = \frac{\sqrt{k(N-k)}}{N}. \quad (29)$$

Hence the exact finite-group half-mass is

$$\nu_N^{\text{GRPO}}(p) = \frac{1}{N} \mathbb{E}_{K \sim \text{Bin}(N,p)} \sqrt{K(N-K)}, \quad (30)$$

and the finite- N population gradient weight is

$$w_N^{\text{GRPO}}(p) = \frac{\nu_N^{\text{GRPO}}(p)}{pq}. \quad (31)$$

As $N \rightarrow \infty$ at fixed $p \in (0, 1)$,

$$\nu_N^{\text{GRPO}}(p) \rightarrow \sqrt{pq}, \quad w_N^{\text{GRPO}}(p) \rightarrow \frac{1}{\sqrt{pq}},$$

recovering the population curve in the MaxRL baseline.

If the implementation uses the sample standard deviation with denominator $N - 1$, multiply eq. (30) by $\sqrt{(N-1)/N}$. State the convention in the paper and code audit.

12.3 Exact tail comparisons

As $p \downarrow 0$, the $K = 1$ event dominates:

$$\nu_N^{\text{GRPO}}(p) = p\sqrt{N-1} + O(p^2). \quad (32)$$

Together with eq. (19),

$$\frac{\nu_N^{\text{MaxRL}}(p)}{\nu_N^{\text{RLOO}}(p)} \rightarrow N-1, \quad \frac{\nu_N^{\text{MaxRL}}(p)}{\nu_N^{\text{GRPO}}(p)} \rightarrow \sqrt{N-1}. \quad (33)$$

As $p \uparrow 1$,

$$\nu_N^{\text{MaxRL}}(p) \sim q, \quad \nu_N^{\text{RLOO}}(p) \sim q, \quad \nu_N^{\text{GRPO}}(p) \sim q\sqrt{N-1}. \quad (34)$$

Therefore MaxRL and RLOO have the same first-order mastered-tail decay; GRPO retains a $\sqrt{N-1}$ -larger coefficient mass under the stated finite-group convention.

Corrected Figure 1 interpretation

At fixed N , practical MaxRL concentrates $(N-1) \times$ more normalized coefficient mass than RLOO on the hard tail and $\sqrt{N-1} \times$ more than finite-group GRPO. In the mastered tail, MaxRL and RLOO both decay linearly, while GRPO retains a $\sqrt{N-1}$ -larger mass.

13 One-step rollout allocation and water-filling

For fixed pass rate p_i and group size $n \geq 1$, define the normalized practical-MaxRL activity

$$U_i(n) = q_i - q_i^n, \quad q_i = 1 - p_i.$$

The marginal activity from adding one rollout to a group of current size n is

$$\Delta_i(n) = U_i(n+1) - U_i(n) = p_i q_i^n. \quad (35)$$

This is exactly the probability that the new rollout is the first success after n failures.

Proposition 13.1 (Discrete one-step mass water-filling). *Suppose each selected task begins with a fixed initial group size $n_i \geq 1$, pass rates remain fixed during allocation, and an additional integer rollout budget is allocated to maximize*

$$\sum_i U_i(n_i).$$

Because $\Delta_i(n) = p_i q_i^n$ is nonincreasing in n , repeatedly allocating the next rollout to the task with the largest current marginal is optimal for this separable one-step activity objective.

Proof. Each U_i is discretely concave because its forward differences $p_i q_i^n$ decrease with n . Maximizing a sum of separable discretely concave utilities under a cardinality constraint is solved by selecting the largest available marginal increments. An exchange argument shows that any allocation containing a smaller chosen marginal while excluding a larger available marginal can be improved by swapping them. $\square$

Evidence boundary

This is a one-step mass oracle, not a universal ceiling on AUC or final held-out performance. Sequential curricula change the policy, transfer across tasks, interfere through shared parameters, update the posterior, and alter future pass rates. Use “mass-optimal allocator under fixed pass rates” rather than “oracle ceiling for pure samplers.”

14 What estimator activity cannot decide: task transfer

The exact activity profile answers whether a task can emit a contrastive update. It does not answer whether that update improves the held-out objective. For evaluation objective $J_\rho(\theta)$ and a small step η , training on task i has first-order value

$$\Delta J_\rho(i) \approx \eta \nabla_\theta J_\rho(\theta)^\top \mathbb{E}[\hat{g}_i]. \quad (36)$$

Using eq. (6),

$$\Delta J_\rho(i) \approx \eta \nu_N(p_i) \nabla_\theta J_\rho(\theta)^\top (\mu_{i,+} - \mu_{i,-}). \quad (37)$$

The first factor is estimator activity. The second is transfer/alignment. A curriculum requires both.

This distinction explains several results already present in the draft:

- shared-policy MountainCar can transfer progress from easier goals to the flag, while per-bin parameters cannot;

- sharper concentration helps structured skill chains where learning compounds, but not flat pools;
- one-shot task streams provide little opportunity for a relabeled skill to compound;
- on learnable-everywhere terrain, breadth can beat frontier focus;
- a narrow template stratum may provide activity without enough rollout-set diversity or transferable structure.

Ready-to-use rewrite

The estimator determines whether a task can emit contrast under the current group budget; shared representation and task geometry determine where that contrast transfers.

15 Hindsight relabeling: a correct sampling-law theorem

The current draft has already moved toward a conditional-law caveat, but the theorem should be made fully explicit.

Fix a destination task g . Let $Q_\theta^{x \rightarrow g}$ denote the proposal law over rewritten trajectories produced by sampling under source task x , selecting destination g , and applying the conditioning rewrite. Let

$$\Pi_\theta^g(\mathrm{d}z) = \pi_\theta(\mathrm{d}z \mid g)$$

be the fresh destination policy law. Define destination verifier $R_g(z) \in \{0, 1\}$ and destination score

$$s_g(z) = \nabla_\theta \log \pi_\theta(z \mid g).$$

When taking the relabel update, treat the sampled proposal as stop-gradient data; the following is a semi-gradient statement.

15.1 Fixed-destination proposal theorem

Theorem 15.1 (Relabeled-group factorization). *Suppose $z_1, \dots, z_N$ are i.i.d. from a fixed proposal $Q^{x \rightarrow g}$, rewards are recomputed as $r_i = R_g(z_i)$, and the practical balanced coefficients are applied to the destination scores $s_g(z_i)$. Define*

$$p_Q = \mathbb{E}_Q[R_g(z)], \quad \mu_{Q,+}^g = \mathbb{E}_Q[s_g(z) \mid R_g = 1], \quad \mu_{Q,-}^g = \mathbb{E}_Q[s_g(z) \mid R_g = 0].$$

Then the expected hindsight semi-gradient is

$$G_H(g) = \nu_N(p_Q) \left(\mu_{Q,+}^g - \mu_{Q,-}^g \right). \quad (38)$$

Proof. The proof is identical to theorem 8.2: conditional on $K = k$, the total positive coefficient is a_k and the total negative coefficient is $-a_k$, but the conditional score means are taken under the proposal Q while the score itself is evaluated under the destination policy. $\square$

The fresh on-policy practical MaxRL update at destination g is

$$G_{\text{on}}(g) = \nu_N(p_\Pi) \left(\mu_{\Pi,+}^g - \mu_{\Pi,-}^g \right) = \frac{1 - (1 - p_\Pi)^{N-1}}{p_\Pi} \nabla_\theta p_\Pi, \quad (39)$$

where $p_\Pi = \mathbb{E}_{\Pi^g}[R_g]$.

Corollary 15.2 (Exact equality condition). *The hindsight and fresh destination updates are equal if and only if*

$$\nu_N(p_Q) (\mu_{Q,+}^g - \mu_{Q,-}^g) = \nu_N(p_\Pi) (\mu_{\Pi,+}^g - \mu_{\Pi,-}^g). \quad (40)$$

A sufficient condition is equality of the proposal and destination laws, $Q^{x \rightarrow g} = \Pi^g$. Matching both conditional laws and the success probability is also sufficient.

Remark 15.3 (Why success-law matching alone is insufficient for the centered estimator). The practical centered estimator uses successful trajectories positively and failed trajectories negatively. Therefore equality of only the success-conditioned laws is insufficient unless the failure contribution also matches or cancels. A raw success-only estimator would require a different equality condition.

Remark 15.4 (All-success relabel groups are silent). If the rewrite makes every trajectory in the group a success, then $p_Q = 1$ and $\nu_N(1) = 0$. The practical centered estimator produces zero update. The paper must specify how the relabeled destination is chosen and how the new success count K' retains contrast.

15.2 A distribution-shift bias bound

Assume $\|s_g(z)\| \leq B$ under all relevant conditional laws. Let Q_+, Q_- and Π_+, Π_- denote success- and failure-conditioned laws. Then

Proposition 15.5 (A simple total-variation bound).

$$\|G_H(g) - G_{\text{on}}(g)\| \leq 2B |\nu_N(p_Q) - \nu_N(p_\Pi)| \quad (41)$$

$$+ 2B \nu_N(p_Q) [\text{TV}(Q_+, \Pi_+) + \text{TV}(Q_-, \Pi_-)]. \quad (42)$$

Proof. Add and subtract $\nu_N(p_Q)(\mu_{\Pi,+}^g - \mu_{\Pi,-}^g)$ and apply the triangle inequality. Since each conditional mean has norm at most B , the target score contrast has norm at most $2B$. For any vector-valued function bounded by B , the difference in expectations under two laws is at most $2B$ times their total variation distance. Apply this separately to success and failure conditional means. $\square$

This bound is diagnostic rather than a guarantee; neural-network scores are not globally bounded unless clipped. It nevertheless makes the required measurements clear.

15.3 Same-group destination selection creates an additional selection issue

If the destination $g = H(z_{1:N})$ is selected from the same group later used for the update—for example, the deepest achieved subgoal—then the trajectories are not generally i.i.d. from a fixed proposal conditional on the selected destination. The cleanest theorem applies to a fixed or cross-fitted destination.

A cross-fitted variant is:

1. split the dead source group into a destination-selection subset and an update subset;
2. choose g using only the selection subset;
3. rewrite and reverify the disjoint update subset at g ;
4. apply the practical estimator to the update subset.

This does not remove source-to-destination off-policy shift, but it removes adaptive reuse of the same trajectories for choosing and estimating the destination update.

Recommended hindsight proposition

A relabeled group yields a verified destination-conditioned semi-gradient under the source-induced proposal law. It equals the fresh destination practical-MaxRL update only when the proposal activity and success–failure score contrast match those of fresh destination rollouts; equality of the full laws is sufficient. Exact verification guarantees semantic correctness, not on-policy destination sampling.

16 Recycling-induced sharpening: what can be formalized

Let $q_t(g)$ be the source-induced frequency with which candidate relabel destinations are produced at training time t , and let $a_t(g) \in [0, 1]$ be the admission rule. The admitted destination marginal is

$$\tilde{q}_t(g) = \frac{q_t(g)a_t(g)}{\sum_h q_t(h)a_t(h)}. \quad (43)$$

Without importance correction to a desired destination distribution $\rho(g)$, the relabel stream tilts training toward $\tilde{q}_t$.

A useful concentration statistic is

$$C(q_t) = \sum_g q_t(g)^2 = \mathbb{E}_{G \sim q_t}[q_t(G)], \quad (44)$$

which is the collision probability. Destinations sampled from the policy’s achieved outcomes are, descriptively, drawn more often from high-occupancy modes.

Evidence boundary

Equation eq. (44) does not by itself prove sharpening. Under perfectly matched, unfiltered self-sampling, $\mathbb{E}_{G \sim q}[\nabla_\theta \log q(G)] = 0$. A nonzero concentration effect must come from curation, cross-task law mismatch, adaptive destination selection, unequal success/failure contrast, finite-sample optimization, or some combination. Treat occupancy bias as a mechanism to test, not a completed proof.

The Countdown finding is still important: mean accuracy rises while pass@16 falls across three seeds. The stronger mechanism section should show which of the following explains it:

1. destination-marginal shift toward a small set of reachable values;
2. off-policy score mismatch at the destination;
3. larger effective gradient dose on a subset of tasks;
4. reduced within-task output support;
5. selection of destinations from the same group used for updating.

High-value controls are destination-frequency-matched fresh rollouts, inverse-propensity weighting, a pass-rate-matched random destination, live-gradient replay, and cross-fitted destination selection.

17 The gate: what is derived and what remains tuned

The endpoint behavior of $\nu_N(p)$ motivates rejecting saturated destinations. It does not uniquely derive the hard threshold used in the current algorithm.

17.1 Hard upper-pass-rate gate

The deployed rule is

$$a_\tau(g) = \mathbf{1}\{\hat{p}_g \leq \tau\}. \quad (45)$$

This is simple and interpretable, but τ is tuned.

17.2 Utility-based gate

A more algebraically faithful alternative is

$$a_\lambda(g) = \mathbf{1}\{\nu_N(\hat{p}_g) \geq \lambda\}, \quad (46)$$

or a soft rule

$$a_\beta(g) = \frac{\nu_N(\hat{p}_g)^\beta}{\max_h \nu_N(\hat{p}_h)^\beta + \varepsilon}. \quad (47)$$

These reject both extremely hard and saturated destinations; whether that is desirable is empirical.

Ready-to-use rewrite

The estimator's endpoint zero motivates a saturation-aware admission rule. We instantiate that principle with a tunable pass-rate threshold; the gate strength, posterior decay, replay dose, concentration exponent, and uniform floor remain implementation parameters.

The current three-seed gate result should be labeled an unintended intermediate operating point because the decay bug changed the effective threshold. The final paper should report a corrected fixed-decay sweep, not reinterpret the bug as if it had been designed.

18 Theory-to-code contract

A paper whose headline is estimator algebra should include a compact audit table connecting assumptions to code.

Theory assumption	Code question	Consequence if different
Binary rewards	Are rewards exactly 0/1 after every verifier and rewrite?	Continuous or shaped rewards invalidate the binomial and coefficient-mass formulas.

Theory assumption	Code question	Consequence if different
I.i.d. rollouts within a group	Are rollouts independent, or coupled by shared decoding state, batching, or diversity constraints?	K need not be binomial; pass@N and mass formulas change.
Exact denominator K/N	Is an epsilon added?	With $\varepsilon > 0$, the estimator is slightly rescaled. For $\varepsilon = 10^{-6}$ and $N \leq 128$, the relative error is at most roughly 1.3×10^{-4} for live groups, but “exact” should still be qualified.
One sequence score per trajectory	Is loss normalized by prompts, trajectories, or total nonpadding tokens?	Total-token normalization introduces response-length-dependent prompt weighting and modifies exact prompt-level objective equivalence.
Fully on-policy requested-task data	Are there multiple optimizer epochs, stale rollout policies, clipping, or importance ratios?	PPO-style clipping and stale data add bias not captured by the theorem.
Destination data sampled from target law	Are relabeled trajectories generated under the destination prompt?	If not, the update is a destination semi-gradient under a proposal law.
Global group statistics	Are mean and standard deviation computed across the full group or per device shard?	Per-shard normalization changes coefficients and exact mass.
GRPO population standard deviation	Does implementation use denominator N or $N - 1$?	Tail constants and finite-group mass differ by $\sqrt{(N - 1)/N}$.
Constant groups set to zero	Does the code retain the full control-variate term at $K = 0$?	The activity profile and objective order change from the practical formulas.
Requested-only posterior updates	Are relabeled successes excluded from the teacher posterior?	Including them estimates proposal/relabel reachability rather than requested-task pass rate and can inflate the posterior.

Part III

Questions to raise before submission

19 Mathematical questions reviewers will ask

Before using the word “exact”

1. What is the random variable whose expectation is exact: coefficient mass, gradient, norm, improvement, or pass-rate change?
2. Under which sampling law is the expectation taken: the requested prompt policy, a source-induced relabel proposal, or the destination policy?
3. Is the score function evaluated under the same law that generated the sample?
4. Is the relabel destination fixed before sampling, selected from an independent group, or chosen from the same trajectories used for updating?

5. Does the implementation add epsilon, clip importance ratios, average by tokens, or use stale rollouts?
6. Are the endpoint statements literal at $p \in \{0, 1\}$ or asymptotic for a full-support softmax model?

Questions to answer in the theory section:

1. Why is coefficient mass the correct pre-rollout quantity rather than live-group probability $\ell_N(p)$, gradient variance, or an expected-improvement proxy?
2. Can two prompts with the same p and $\nu_N(p)$ have opposite transfer to held-out performance? If so, what exactly does the curriculum theorem promise?
3. Does the water-filling proposition allocate total rollouts across prompts, or only additional rollouts after each selected prompt has a mandatory group?
4. What is the exact definition of the learnable band’s width? If there is no threshold, should the paper claim only the profile and peak are derived?
5. Which MaxRL estimator is used in every theorem and experiment: raw, full control variate, or practical drop-all-fail?
6. What is the coefficient convention for RLOO and the standard-deviation convention for GRPO?
7. Does the practical $T = N - 1$ result survive the actual token-normalized LLM loss, or only the sequence-level idealization?
8. If all relabeled trajectories succeed at the new destination, why does the centered estimator not vanish? Where does relabeled contrast come from in code?
9. Is the gate meant to reject only saturated destinations, or both hard and saturated destinations? Which objective derives that choice?
10. Is the “frontier” defined by p_N^* , a level set of ν_N , a posterior quantile, or an empirical task stratum?

20 Causal identification questions

Before claiming the curriculum harms or helps an estimator

1. Did MaxRL and GRPO receive the same realized prompt sequence, or only the same adaptive teacher rule?
2. If schedules differ, is the claim about the estimator alone or the full estimator–teacher feedback loop?
3. Was the treatment actually delivered? How far did the sampled task distribution move from uniform?
4. Are replacement, prompt repetition, early termination, skipped backward passes, and response length alternative explanations?

5. Are wall-clock, optimizer steps, generated tokens, and verifier calls all matched or separately reported?

For the maze and GSM8K studies, define the interaction estimand explicitly. For paired seed or warmstart s , let

$$Y_{E,C,s}$$

be the outcome for estimator $E \in \{M, G\}$ (MaxRL or GRPO) and curriculum $C \in \{T, U\}$ (teacher or uniform). The curriculum-by-estimator interaction is

$$D_s = (Y_{M,T,s} - Y_{M,U,s}) - (Y_{G,T,s} - Y_{G,U,s}). \quad (48)$$

A coverage main effect of estimator is a different estimand. Do not substitute one for the other.

Questions to answer:

1. Is the primary endpoint final pass@k, change from warmstart, AUC, or multi-sample coverage gain? Was it pre-declared?
2. Is the teacher’s effect estimated relative to uniform-without-replacement, uniform-with-replacement, or a fixed external schedule?
3. Does the teacher change only task identity, or also sequence length and the number of optimizer steps per hour?
4. If recycling fills otherwise empty update slots, is the gain label information, gradient dose, or both? Which placebo isolates each component?
5. Does a relabel improve the destination task, the source task, neighboring tasks, or only the aggregate mean? Where is transfer measured?
6. Does the gate recover coverage because it rejects saturated destinations, because it lowers total dose, or because it changes destination diversity?

21 Statistical questions

The exact one-sided permutation value 0.0079 is recognizable: with five labels of one kind and four of another, complete separation gives

$$p_{\min} = \binom{9}{4}^{-1} = \frac{1}{126} \approx 0.00794. \quad (49)$$

That calculation is valid only if the nine observations are exchangeable under the null.

Before reporting a small exact p -value

1. Are the nine observations independent training runs, or do several share seeds, warmstarts, schedules, or evaluation samples?
2. Why do five MaxRL observations enter one test while eighteen enter another?
3. Were all eligible configurations and exclusion rules declared before inspecting the outcome?

4. Are two teacher variants within one random seed being counted as two independent seeds?
5. Is the permutation blocked within warmstart or seed clusters?
6. Are multiple metrics and difficulty bands being tested without a declared primary contrast?

Recommended reporting:

1. provide a public run registry with one row per training run;
2. plot every paired seed/warmstart contrast;
3. use blocked randomization or cluster bootstrap at the warmstart/seed level;
4. call six variant-by-seed comparisons “six paired contrasts within three random seeds,” not six seeds;
5. report effect sizes and confidence intervals even when an exact sign pattern is striking;
6. distinguish confirmatory runs from exploratory variants and post-hoc analyses.

22 Implementation and reproducibility questions

1. Can a reader reconstruct every coefficient used by MaxRL, GRPO, and RLOO from the paper without opening code?
2. Are group statistics computed before or after distributed sharding?
3. Is the policy update averaged per prompt, per trajectory, or per token?
4. Are relabeled prompts and responses both rewritten? Are verifier inputs and training inputs identical after rewrite?
5. Is the relabeled success count K' recomputed for the whole group?
6. Does the posterior receive only requested-task evidence? How are decay and checkpoint restoration implemented?
7. Are gate keys unique to relabeled tasks, or can destinations collide by value across templates?
8. Are pass@k estimates computed by the same harness across trainer validation and final evaluation? The current draft discloses a roughly $3\times$ absolute discrepancy; that must be reconciled or isolated.
9. Are all figures generated from immutable JSON artifacts with a manifest mapping run IDs to panels?
10. Does the anonymous artifact contain usernames, absolute paths, Git remotes, experiment-tracking URLs, or PDF metadata?

23 Mechanism questions for recycling-induced sharpening

What would falsify the proposed mechanism?

1. If relabel destinations are frequency-matched but generated freshly under the destination prompt, does coverage still fall?
2. If relabel dose is matched with live-gradient replay, how much loss remains?
3. If inverse destination-frequency weighting is used, does pass@k recover?
4. If the destination is selected from a disjoint group, does the effect shrink?
5. If destination pass rate is held fixed while occupancy varies, which variable predicts the damage?
6. Does the same mean-coverage trade occur with raw MaxRL, practical MaxRL, full-CV MaxRL, and GRPO?
7. Is coverage loss within tasks, across task strata, or both?

Telemetry that would make the mechanism persuasive:

- histograms of requested-task and destination $\hat{p}$ over time;
- destination occupancy, collision probability, and effective support;
- admitted versus rejected destination distributions;
- cumulative relabel dose and the time ordering of mean gain versus coverage loss;
- fresh-destination versus relabeled-destination score statistics;
- per-tier pass@1, pass@k, and output-support summaries.

24 Writing questions for every paragraph

Before keeping a sentence, ask:

1. Is this a theorem, an empirical result, a mechanism hypothesis, or an interpretation? Does the verb reveal which one it is?
2. Does “same curriculum” mean the same algorithm or the same realized schedule?
3. Does “safe” mean no reduction in pass@k at one k , no reduction in area under the coverage curve, or no held-out regression at all?
4. Does “coverage” refer to pass@k, summed pass@k over levels, the number of prompts solved, or output diversity?
5. Is the numerical claim paired with its seed count, uncertainty, compute currency, and primary endpoint?

6. Is a negative result diagnosed by evidence, or rationalized after the fact?
7. Does the paragraph make a broader claim than the table or figure actually shows?
8. Can one slogan be removed without losing information?
9. Is the result load-bearing enough for the main paper, or should it be an appendix boundary condition?
10. Would a skeptical reviewer know exactly what experiment would overturn the statement?

Part IV

Complete rewriting guide

25 Title and positioning

The current title is memorable, but “what curricula and failure recycling can and cannot do” sounds universal. Three options are:

1. **Keep the current title, narrow the subtitle in the abstract:** *The Estimator Decides: Finite-Group Curricula and Failure Recycling in RL with Verifiable Rewards.*
2. **More technical:** *Finite-Group Estimator Activity for Curricula and Hindsight Relabeling in RLVR.*
3. **Best balance of hook and scope:** *The Estimator Decides Where a Curriculum Can Act.*

The paper should position itself as a complement to MaxRL, not a competitor:

Ready-to-use rewrite

Tajwar et al. characterize the population objective induced by success-conditioned normalization. We characterize the finite-group activity of the practical centered implementation and use that activity before rollout generation, where a curriculum must decide which prompts receive compute.

26 Recommended abstract for the current evidence

Conservative abstract—ready to paste

Difficulty curricula and hindsight relabeling are often treated as objective-agnostic modules for reinforcement learning with verifiable binary rewards. We show that their effect depends on the group estimator underneath. For the practical centered MaxRL estimator that zeros constant-reward groups, a prompt with pass rate p and N rollouts has expected coefficient mass

$$A_N(p) = 2[1 - (1 - p)^N - p].$$

Its normalized half-mass $\nu_N = A_N/2$ also factorizes the expected update as

$$\mathbb{E}[\hat{g} \mid x] = \nu_N(p)(\mu_+(x) - \mu_-(x)),$$

so the curve is exactly the estimator-controlled scalar multiplying the prompt’s success–failure score contrast. The profile has a unique interior maximum and vanishes at $p = 0$ and $p = 1$, motivating a prompt sampler that tracks the moving frontier without hand-setting a target difficulty. The same analysis exposes a limit: with this practical estimator held fixed, reallocating prompts cannot create contrast in an all-fail group. When an exact relabel map exists, hindsight can convert failed source rollouts into verified destination examples, but those examples are generally off-policy for the destination and can tilt training toward already-reachable outcomes. Across maze curriculum variants, MaxRL runs increase pass@8 while GRPO runs decrease it. On Countdown arithmetic, relabeling raises mean success but lowers pass@16 across three seeds; a saturation-aware destination gate recovers frontier coverage at an unintended intermediate operating point while retaining part of the mean gain. A boundary suite shows that the benefits require a graded task pool and transferable shared parameters.

This version deliberately does not put the preliminary LLM interaction, $11\times$ inference threshold, or corrected one-seed full-strength gate in the headline.

27 Stronger abstract after the missing experiments

Use the following only after schedule matching, corrected multi-seed gating, and stronger LLM steering are complete.

Post-replication abstract template

Difficulty curricula and failure recycling are usually evaluated independently of the objective and estimator that consume them. We derive their finite-group interaction for success-conditioned RL. For the practical centered MaxRL estimator, the expected coefficient mass of a prompt with pass rate p and N rollouts is $2(\text{pass}@N - \text{pass}@1)$, and the normalized mass is exactly the scalar multiplying the success–failure score contrast. This yields a compute-indexed frontier sampler and predicts where reallocation is silent. We then characterize relabeling as a verified destination semi-gradient and show that destination saturation creates a mean–coverage trade. In schedule-matched experiments, the same prompt sequence preserves pass@k under MaxRL but reduces it under GRPO. In corrected multi-seed Countdown runs, a saturation-aware gate traces a monotone mean-versus-coverage frontier. Across controlled chains, mazes, and LLM reasoning tasks, allocation improves efficiency where a graded frontier exists, while relabeling is necessary when the requested pool is operationally dead. These results show that estimator activity constrains where curricula can act and that relabel destination selection is a first-class coverage decision.

28 Recommended introduction opening

Opening two paragraphs—ready to paste

Post-training with binary verifiers is constrained by a simple event: a practical group-relative update needs reward contrast. If all N rollouts fail or all N succeed, the centered estimator is silent. This event-level constraint is not visible in population weight functions, which describe how a prompt is weighted after averaging over sampling. A curriculum acts one step earlier, deciding

which prompts receive the rollout budget that makes those weights observable at all. MaxRL shows that likelihood-style objectives place more population weight on hard prompts after they are sampled. We ask the complementary finite-group question: given a rollout budget N , which prompts can emit a contrastive update before that budget is spent? For the practical centered MaxRL estimator, the answer is closed form. Its normalized expected coefficient mass is $\text{pass}@N(p) - \text{pass}@1(p)$, and this quantity exactly multiplies the prompt’s success–failure score contrast. The result gives a principled allocation profile, clarifies the implementation-level gap between raw and practical MaxRL, and exposes what prompt reallocation cannot do.

Then introduce relabeling:

Ready-to-use rewrite

The same boundary motivates a complementary intervention. When a requested-task group is all-fail, reallocation cannot recover the compute already spent. If the environment provides an exact relabel map, the trajectories may still form a verified group for an achieved destination. This creates contrast, but it also changes the destination distribution and generally produces off-policy destination data. We therefore study allocation and relabeling as separate channels: one decides where existing contrast is likely, the other changes which task the data train.

29 Short contribution list

Four contributions—ready to paste

1. **Finite-group theory.** We distinguish the raw, full-control-variate, and practical MaxRL estimators and derive the practical estimator’s exact coefficient-mass profile and $T = N - 1$ objective order.
2. **Estimator-derived allocation.** We use the normalized profile as a prompt-sampling utility and show that its $N = 2$ slice recovers the standard learnability curve, while its finite- N comparison with GRPO predicts different hard- and mastered-tail activity.
3. **Relabeling characterization and coverage cost.** We characterize hindsight as a verified but generally off-policy destination semi-gradient and identify recycling-induced sharpening: mean success can rise while $\text{pass}@k$ coverage falls.
4. **Admission and boundaries.** A destination-saturation gate traces a mean-versus-coverage trade, while negative suites identify the conditions required for benefit: a graded pool, shared transfer, and reusable relabeled skills.

30 Recommended paper structure

30.1 1. Introduction

One problem, one theorem preview, three empirical claims, and one explicit evidence boundary.

30.2 2. From MaxRL’s population weight to finite-group activity

- notation and binary verifier setup;
- raw, full-CV, and practical estimators in one table;
- conditional score identities;
- coefficient-mass factorization;
- practical $T = N - 1$ corollary.

30.3 3. Estimator-derived allocation

- ν_N , exact peak, tails, and operational budget definition;
- RLOO and finite-group GRPO comparison;
- Thompson posterior sampler and optional one-step water-filling;
- explicit limitation: activity does not equal transfer.

30.4 4. Failure recycling and destination admission

- exact verifier and conditioning-rewrite contracts;
- source proposal versus destination policy law;
- relabeled-group factorization and equality condition;
- destination-marginal shift and gate.

30.5 5. Experiments

1. **Exact chains and frontier-heavy attribution:** utility, full-CV baseline, replay placebo, allocation versus creation.
2. **Schedule-matched maze study:** estimator coverage effect, level/time mechanism, wall-clock versus token/step decomposition.
3. **Countdown sharpening and corrected gate:** mean-coverage trade, destination telemetry, multi-seed sweep.
4. **Boundary synthesis:** Gym shared-parameter ablation, Jugs, IsaacLab, and one-shot streams in one table.

30.6 6. Related work, limitations, and conclusion

Move preliminary GSM8K, single-checkpoint inference thresholds, detailed classic-control curves, and most audit history to appendices until their evidence is stronger.

31 Theorem statements ready for the manuscript

Lemma: practical objective order

Lemma (Objective order of the practical estimator). Let K be the number of successes among N i.i.d. binary-reward rollouts. The raw success-conditioned estimator is unbiased for the order- N MaxRL objective. If the unconditional average-score control variate is subtracted but the entire group is zeroed when $K = 0$, the resulting practical estimator is unbiased for order $N - 1$.

Proposition: coefficient mass and factorization

Proposition (Finite-group activity). For the practical centered estimator,

$$A_N(p) = \mathbb{E} \sum_i |w_i| = 2[(1 - p) - (1 - p)^N].$$

Writing $\nu_N = A_N/2$, the expected update satisfies

$$\mathbb{E}[\hat{g}_N \mid x] = \nu_N(p) (\mathbb{E}[S \mid r = 1, x] - \mathbb{E}[S \mid r = 0, x]).$$

Thus ν_N is the estimator-controlled scalar multiplying the prompt-specific success–failure score contrast.

Proposition: peak

Proposition (Compute-indexed activity peak). The function $\nu_N(p) = (1 - p) - (1 - p)^N$ is strictly concave on $(0, 1)$ and has the unique maximizer $p_N^* = 1 - N^{-1/(N-1)}$. Its hard-tail expansion is $(N - 1)p + O(p^2)$ and its mastered-tail expansion is $(1 - p) + o(1 - p)$.

Proposition: relabeled update

Proposition (Relabeled destination semi-gradient). Fix a destination g and let $Q^{x \rightarrow g}$ be the source-induced proposal over rewritten trajectories. Applying the practical centered coefficients to destination scores yields

$$G_H(g) = \nu_N(p_Q)(\mu_{Q,+}^g - \mu_{Q,-}^g).$$

This equals the fresh destination update only when the proposal activity and conditional score contrast match those of rollouts sampled from $\pi_\theta(\cdot \mid g)$. Exact verification guarantees semantic correctness of the relabeled reward, not equality of the sampling laws.

32 Section-by-section rewriting guidance

32.1 Abstract

Remove or qualify:

- “expected learning signal” $\rightarrow$ “expected coefficient mass”;
- “recycling is the only channel” $\rightarrow$ “with the practical zero-constant-group estimator fixed, reallocation cannot create contrast in an all-fail group”;

- “the same curriculum degrades GRPO” $\rightarrow$ “coverage behavior separates by estimator across curriculum variants”;
- “teacher-deficit direction replicates 2/2 seeds” $\rightarrow$ “the endpoint contrast has the same sign in two seeds, while the pre-registered regression shape appears in one”;
- “gate restores coverage” $\rightarrow$ state the unintended operating point and corrected rerun status;
- “pre-registered fifth suite defeats every estimator” $\rightarrow$ explain the measured boundary: narrow template support and insufficient rollout diversity at the deployed N .

32.2 Introduction

The current introduction has three strong questions, but the prose repeats the partition slogan before the estimator distinctions are established. Reorder:

1. constant-group event and compute waste;
2. MaxRL population weight versus finite-group activity;
3. exact practical estimator result;
4. allocation channel;
5. relabel channel and off-policy destination issue;
6. coverage metric and empirical questions;
7. contributions.

Do not introduce “safe” until the metric is defined. Prefer “coverage-stable on held-out pass@k.”

32.3 Theory section

Put the three-estimator table before the $T = N - 1$ lemma. This prevents a reviewer from reading the lemma as a contradiction of the baseline theorem. Follow every theorem with two interpretation sentences:

1. what is exact;
2. what is not implied.

For Proposition 1, use A_N and ν_N consistently. Replace “the estimator’s expected learning signal” with the factorization interpretation.

For the comparison proposition, correct the mastered-tail statement: MaxRL and RLOO decay at the same first order; GRPO retains the larger mass.

For water-filling, call it a fixed-pass-rate one-step mass oracle and state the minimum group-size assumption.

32.4 FrontierMax method section

Clarify the deployed sampler and the optional allocator as different algorithms. The practical sampler uses fixed N and samples prompts according to a posterior-derived utility. The water-filling oracle changes N_i and therefore changes both activity and objective order.

The algorithm should explicitly show:

1. requested prompt x ;
2. requested-task success count K ;
3. destination g' selected from a dead group;
4. rewritten prompt and response;
5. reverified relabeled success count K' ;
6. admission based on destination evidence;
7. posterior updated only with requested-task outcomes.

32.5 Failure recycling section

Replace the headline “Hindsight gradients are exact, not approximate” with the proposal-law theorem. Keep the placebo decomposition, because it is one of the draft’s best pieces of scientific honesty:

Ready-to-use rewrite

Most of recycling’s fixed-pool AUC gain is explained by filling otherwise-unused update slots with additional gradient dose; the relabel direction contributes a smaller but reproducible increment beyond the strongest replay placebo. We therefore report dose and direction as separate effects.

Do not use cosine probes as primary evidence if null relabeling schemes produce the same cosine. The behavioral placebo battery is stronger.

32.6 Maze section

Lead with what the design establishes:

Ready-to-use rewrite

Across selected curriculum and recycling variants, held-out pass@8 changes separate perfectly by estimator: the MaxRL variants increase coverage and the GRPO variants decrease it. This is an estimator-conditioned coverage main effect robust to the data-level variants tested. Whether the teacher itself amplifies GRPO’s decline is a separate factorial interaction question.

Then show the level/time mechanism. Rename “diversity premium” to “multi-sample coverage gain” or “sampling premium.” Explain the $n = 5$ versus $n = 18$ inclusion rules in the paper, not only the repository.

32.7 GSM8K section

The most interesting current result is posterior starvation, not a confirmed interaction. Rewrite the section around treatment delivery:

Ready-to-use rewrite

The prompt-level posterior learns a weak but real difficulty ordering, yet the rollout budget visits too few of 7,473 prompts for the posterior to become decisive. As a result, the delivered sampling distribution remains near uniform. The registered GRPO teacher deficit has the same endpoint sign in two seeds but the pre-registered regression trajectory appears in one; magnitude tracks the small amount of measured steering. We therefore treat this experiment as a diagnosis of posterior starvation and as motivation for a stronger, steering-controlled replication.

Disclose the replacement confound next to the result, not in a later caveat.

32.8 Countdown and gate sections

Split saturated and frontier tiers. At tier 1, the moderate gate mostly neutralizes recycling. At the frontier tier, it reportedly retains about 60% of the mean gain while restoring coverage. Do not place the frontier claim only in a tier-1 caption.

Use this wording for the bug:

Ready-to-use rewrite

A decay implementation error made the three-seed gate weaker than intended and produced an unintended intermediate operating point. We report it because it is informative about the tradeoff, but the final claim is based on a fixed-decay sweep rather than treating the bugged setting as designed.

32.9 Limitations

The limitation section is already strong. Make it more synthetic by grouping failures into necessary conditions:

Table 5: Recommended boundary-condition synthesis.

Required condition	Failure when absent	Evidence in the draft
Nonconstant group probability at budget	target-only group-relative training freezes	sparse Gym official tasks
Shared transferable parameters	easier goals do not unlock harder goals	per-bin policy ablation
Graded support over difficulty	no frontier exists for the teacher to walk	Jugs template stratum
Meaningful avoidable dead region	breadth can beat focus	IsaacLab pilot
Reusable relabeled skills	relabel gains do not compound	one-shot stream versus fixed pool
Unsaturated destinations	relabel dose concentrates mean at coverage’s expense	Countdown
Correct task keying and rewrite	gate collisions or semantically invalid updates	Jugs key bug and gridworld rewrite ablation

33 Sentence-level replacement table

Current phrasing	Recommended replacement
“The estimator learns only from successes.”	“The practical estimator is success-gated: a group contributes only after at least one success appears; within a live group, successes receive positive coefficients and failures negative coefficients.”
“The expected learning signal is exactly $2(\text{pass}@N - \text{pass}@1)$.”	“The expected coefficient mass is exactly $2(\text{pass}@N - \text{pass}@1)$, and its half-mass is the scalar multiplying the success–failure score contrast.”
“The two zeros partition difficulty into a dead zone, a learnable band, and a mastered tail.”	“The endpoint zeros and unique interior maximum define three operational regimes at a fixed budget: negligible contrast, high estimator activity near the frontier, and a saturated tail.”
“No sampler can reach the dead zone.”	“At the deployed (B, N) budget, a low-pass-rate task can be operationally dead for the practical zero-constant-group estimator.”
“Recycling is the only channel into the dead zone.”	“With the practical estimator held fixed, prompt reallocation cannot manufacture contrast in an all-fail group; relabeling is the signal-creation mechanism studied here.”
“MaxRL releases mastered prompts fastest.”	“MaxRL and RLOO both decay linearly in the mastered tail, while finite-group GRPO retains a $\sqrt{N-1}$ -larger coefficient mass.”
“The same curriculum degrades GRPO.”	“Coverage behavior separates by estimator across curriculum variants; the teacher interaction is tested separately.”
“A failed trajectory is a verified success for the goal it reached.”	“An exact relabel map can convert a failed source trajectory into a semantically valid example for an achieved destination, although the destination data are generally off-policy.”
“Hindsight gradients are exact, not approximate.”	“Hindsight yields a verified destination semi-gradient; it equals the fresh destination update under a conditional-law matching condition.”
“Perfect difficulty information is worth almost nothing.”	“The mass oracle improves over posterior-only allocation, while recycling allows the posterior-based stack to approach the oracle’s empirical level.”

Current phrasing	Recommended replacement
“The gate is derived from the same algebra.”	“The endpoint algebra motivates a saturation-aware gate; its operating threshold remains a tunable coverage–mean tradeoff parameter.”
“The bug favorably sampled the middle.”	“The bug produced an unintended intermediate effective gate strength; corrected multi-seed sweeps are required for the final claim.”
“Diversity premium.”	“Multi-sample coverage gain,” “sampling premium,” or simply $\text{pass}@k - \text{pass}@1$.
“Environment-agnostic framework.”	“Environment-independent core with domain-specific task, verifier, and relabel adapters.”

34 Figure and caption guidance

34.1 Figure 1: estimator activity curves

Keep the figure, but make these changes:

1. label the y-axis “normalized expected coefficient mass ν_N ”;
2. state “practical MaxRL, group size N , objective order $T = N - 1$ ”;
3. use A_N for full mass and ν_N for half-mass consistently;
4. correct the mastered-tail comparison with RLOO;
5. distinguish finite-group GRPO from the population GRPO weight curve;
6. use a logit axis or tail insets so both endpoints are visible;
7. make the caption state the standard-deviation convention.

Suggested caption:

Ready-to-use rewrite

Finite-group estimator activity. Left: $\nu_N(p) = (1 - p) - (1 - p)^N$ for the practical centered MaxRL estimator. The exact peak $p_N^* = 1 - N^{-1/(N-1)}$ shifts toward harder prompts as N grows. Right: normalized expected coefficient mass at $N = 16$ for practical MaxRL, RLOO, and finite-group GRPO under population-standard-deviation normalization. MaxRL amplifies the hard tail relative to both baselines; MaxRL and RLOO share the same first-order mastered-tail decay, while GRPO retains a $\sqrt{N - 1}$ -larger mass.

34.2 Method diagram

The method diagram should show the source and destination laws explicitly. Add labels for source prompt x , destination g' , rewrite, re-verification, relabeled success count K' , and destination gate. Do not imply that every trajectory becomes a success; the centered estimator needs relabeled contrast.

34.3 Maze mechanism figure

This is the strongest empirical figure. Keep it in the main paper, but:

- show paired seed/warmstart units rather than unstructured run dots;
- explain $n = 18$ versus $n = 4$ in the caption;
- rename the sampling premium;
- use cluster-aware intervals;
- separate the estimator main effect from teacher interaction language.

34.4 Countdown figure

Use separate saturated-tier and frontier-tier panels. Each should show no recycling, ungated, and at least two corrected gate strengths. Add destination pass-rate distributions beneath the outcome plane so the gate mechanism is visible rather than inferred.

34.5 Appendix candidates

Move the preliminary GSM8K four-cell bar chart, classic-control convergence curves, and single-checkpoint inference multiplier to the appendix unless the corresponding evidence is strengthened.

35 What to borrow from the baseline MaxRL paper’s writing

The MaxRL baseline is effective because it separates objective, estimator, variance reduction, and implementation; gives one memorable algebraic change; uses a population weight-function view; and escalates experiments in four clean stages. Your paper should mirror that discipline one level earlier in the pipeline:

1. distinguish the three estimator variants;
2. derive finite-group activity;
3. connect activity to the exact finite- N weight through the factorization theorem;
4. use the profile before sampling;
5. characterize relabeling under its true proposal law;
6. test coverage with matched schedules and destination telemetry.

The openings left by the baseline are sufficient for novelty positioning:

- it does not distinguish the raw $T = N$ theorem from the practical $T = N - 1$ drop-all-fail algorithm;
- its GRPO analysis is population-level, not finite-group constant-event analysis;

- it does not study adaptive prompt allocation;
- it conjectures easy-tail sharpening but does not replay identical schedules across estimators;
- it does not study exact-verifier relabeling or destination-induced pass@k loss.

Do not claim that one scalar fully determines curriculum quality. Claim that it exactly determines the estimator-controlled activity component.

Part V

Experimental and statistical completion plan

36 Decisive experiment 1: full control variate versus practical drop-all-fail

36.1 Question

Is relabeling uniquely necessary because all-fail groups are silent, or because a verified positive anchor is empirically better than the noisy all-fail control-variate update?

36.2 Arms

At minimum:

1. practical MaxRL, zero the full $K = 0$ group;
2. full-CV MaxRL, retain $-N^{-1} \sum_i S_i$ at $K = 0$;
3. practical MaxRL plus recycling;
4. full-CV MaxRL plus recycling, if stable;
5. an optional negative-only or entropy control matched for update count.

36.3 Suites

Run first on the exact chain, the $p \leq 10^{-5}$ frontier-heavy regime, and one maze configuration. These three settings answer complementary questions: mathematical behavior, ignition from an operationally dead pool, and neural-gradient stability.

36.4 Measurements

Report:

- conditional gradient norm and variance on $K = 0$ groups;

- alignment with the exact target gradient where available;
- time to first natural success;
- final/AUC pass rate and pass@k;
- optimizer stability and policy entropy;
- fraction of total update norm contributed by all-fail groups.

36.5 Interpretation

If full-CV fails while recycling succeeds, the paper can say that a nonzero all-fail update is not enough; a verified positive anchor is empirically important. If full-CV works, revise “creation” to include control-variate-based all-fail learning and position relabeling as one robust mechanism rather than the only one.

37 Decisive experiment 2: schedule-matched estimator comparison

37.1 Question

Does coverage separate because of the estimator itself, or because adaptive teachers generate different schedules as policies diverge?

37.2 Design

Use two phases:

1. **Frozen schedule.** Generate prompt IDs, order, and group sizes from an external oracle, a frozen warmstart posterior, or a previously recorded teacher trajectory. Replay the exact schedule under MaxRL and GRPO with paired warmstarts.
2. **Adaptive feedback.** Restore each estimator’s own teacher posterior and report the full deployed loop.

The frozen phase identifies the estimator under a controlled data schedule. The adaptive phase measures the system-level interaction.

37.3 Primary endpoints

Predeclare:

- held-out pass@1;
- held-out pass@8 or the relevant pass@k;
- multi-sample coverage gain $\text{pass}@k - \text{pass}@1$;
- easy/mid and frontier band changes;
- area under the pass@k trajectory.

37.4 Analysis

For each paired seed s , report the estimator contrast under the same schedule. For the adaptive factorial study, report the interaction D_s in eq. (48). With three to five seeds, show raw paired values and use blocked randomization rather than relying on asymptotic normality.

38 Decisive experiment 3: corrected gate sweep

38.1 Question

Does destination saturation, rather than merely lower total relabel dose, explain the mean-coverage trade?

38.2 Arms

Use no recycling, ungated recycling, and at least three gate strengths under fixed, corrected decay. Include both the current hard upper-pass-rate gate and, if inexpensive, one utility-based gate.

38.3 Predeclared tier and metric

Choose one frontier tier as primary before looking at results. Keep the saturated tier as a mechanism contrast. Declare whether the primary utility is mean@16, pass@16, or a scalarized combination.

38.4 Telemetry

Record:

- effective threshold over time;
- destination $\hat{p}$ before admission;
- accepted and rejected destination counts;
- cumulative relabel dose;
- destination collision probability and effective support;
- actor entropy and response/output support;
- mean and coverage trajectories at every tier.

38.5 Required wording

Until this is complete, call the three-seed result an unintended intermediate operating point and the corrected one-seed result a directional follow-up.

39 Decisive experiment 4: LLM treatment delivery

The current GSM8K result is posterior-starved because 3,200 group draws are spread over 7,473 prompts. A stronger test should reduce the teacher’s statistical resolution problem.

39.1 Recommended designs

1. bucket prompts by a stable external difficulty estimate and learn a posterior over buckets;
2. use a kernel or hierarchical posterior over prompt features;
3. increase revisit rate by using a smaller fixed prompt pool;
4. pretrain the posterior on a short uniform warmup, then freeze it for the estimator comparison;
5. add uniform-with-replacement as the direct repetition control.

39.2 Delivered steering metrics

Report more than the dead-sampled fraction. Useful metrics are

$$\text{TV}(q_t, \rho) = \frac{1}{2} \sum_x |q_t(x) - \rho(x)|, \quad (50)$$

$$\text{ESS}(q_t) = \frac{1}{\sum_x q_t(x)^2}, \quad (51)$$

and the cumulative steering dose

$$\mathcal{D}_T = \sum_{t=1}^T \text{TV}(q_t, \rho). \quad (52)$$

A dose-response analysis is meaningful only after the treatment is visibly separated from uniform.

40 High-value mechanism experiments

40.1 Destination-law shift

For a set of relabeled destinations, compare rewritten source trajectories against fresh destination rollouts using:

- a two-sample classifier;
- maximum mean discrepancy;
- response-length and verifier-path summaries;
- score-mean and score-covariance differences;
- success- and failure-conditioned comparisons separately.

This directly operationalizes eq. (42).

40.2 Cross-fitted relabel selection

Compare same-group deepest-achieved relabeling with cross-fitted selection. If the effect changes, adaptive destination reuse is part of the mechanism.

40.3 Transfer matrix

Estimate a local transfer matrix by applying a small update from task or tier i and measuring short-horizon pass-rate changes on tier j . This distinguishes estimator activity from cross-task transfer and can explain why concentration helps chains but not flat pools.

40.4 Generated-token matching

For maze and LLM studies, report outcomes against:

1. wall-clock;
2. optimizer steps;
3. generated tokens or environment transitions;
4. verifier calls;
5. successful/live groups.

A wall-clock-only gain is still valuable, but it is a systems-throughput result rather than a pure allocation-efficiency result.

41 Run registry and statistical analysis

41.1 Required run registry

Use one row per independent training run:

Run ID	Suite	Estimator	Teacher	Recycler	Gate	Seed	Warmstart	Schedule	Confirmatory role
<i>Populate directly from the experiment manifest.</i>									

Add columns for inclusion in each figure/test, code commit, artifact path, wall-clock budget, step count, generated tokens, and evaluation harness version in the actual repository table.

41.2 Blocked permutation

If treatments are paired within warmstart blocks, permute treatment labels only within each block. Do not permute individual configuration runs freely when they share a warmstart or seed.

41.3 Cluster bootstrap

Bootstrap warmstart/seed clusters, carrying all configurations within a selected cluster together. This preserves within-cluster dependence and gives uncertainty for aggregate effects.

41.4 Small- n reporting

With three seeds, prioritize:

- raw paired lines;
- all seed-level contrasts;
- median and mean effect;
- exact sign count;
- sensitivity to leaving one seed out;
- a clearly labeled exploratory p -value if used.

Avoid presenting multiple configuration contrasts as independent replication merely because there are more plotted dots.

42 Main-paper evidence map

A compact main paper can be built around the following evidence chain:

Table 8: Recommended main-paper evidence map.

Rung	Question	Decisive evidence	Claim earned
Theory	What does the practical estimator do at finite N ?	three-estimator distinction, factorization, tails	exact activity profile and implementation-level $T = N - 1$
Chains/frontier	What can allocation versus relabeling contribute?	mass oracle, replay placebo, full-CV baseline, zero-to-solved suite	allocation and signal creation are separable channels
Schedule-matched maze	Does estimator choice change coverage under the same data?	paired frozen schedule and level/time decomposition	estimator-conditioned coverage effect
Countdown	What can relabeling cost, and can admission control it?	replicated mean-up/coverage-down effect and corrected gate sweep	destination admission is a coverage decision
Boundary table	When should the method not work?	shared-parameter, graded-pool, breadth, and one-shot nulls	explicit applicability conditions

43 What belongs in the appendix

Unless strengthened, place these in appendices:

- preliminary GSM8K interaction and pass@k sweep;
- single-checkpoint $11\times$ inference threshold;
- detailed classic-control convergence and dense-reward controls;
- one-seed IsaacLab pilot;
- full audit and retraction timeline;
- every secondary utility variant and concentration exponent;
- detailed hyperparameter sweeps;
- code integration snippets and adapter API;
- extra negative results whose role is boundary diagnosis rather than the main claim.

Part VI

Page-specific audit of the current working draft

44 Page 1: title, abstract, and anonymity

Priority

The manuscript is labeled anonymous but links directly to a repository under a personal username. Remove or replace this before any anonymous submission. Also inspect PDF metadata and supplementary files.

Abstract changes:

- replace expected advantage “learning signal” with coefficient mass and the factorization statement;
- scope the dead zone to the practical estimator and fixed budget;
- remove “recycling is the only channel” until full-CV is tested;
- replace the nine-run result with the accurate estimator main effect;
- do not say the teacher deficit replicates in the same sense as the pre-registered regression shape;
- identify the gate’s current three-seed setting as unintended;
- keep the boundary suite, but define the missing condition as graded support/rollout diversity rather than saying the band simply “collapses.”

45 Page 2: research questions and contributions

Q1 is strong but should say mass is the part of the expected update controlled by the estimator before sampling, not the whole signal. Q2 should define “safe” as held-out coverage-stable. Q3 should state that the gate is motivated by saturation and tuned empirically.

Contribution 2 currently says relabeling yields the ML gradient of the achieved task under the original sampling law. Replace it with the proposal-law semi-gradient theorem. Contribution 3 should not claim confirmation across four suites unless every suite directly tests the same estimator-conditioned prediction.

46 Page 3: Figure 1, Lemma 1, and Proposition 1

Required corrections:

1. fix u_N versus $2u_N$ notation;
2. label the practical estimator and $T = N - 1$ in the figure;
3. distinguish finite-group GRPO mass from population GRPO weight;
4. replace “MaxRL releases mastered prompts fastest” with the correct MaxRL/RLOO/GRPO tail comparison;
5. replace Proposition 1’s interpretation with the factorization theorem;
6. retain the exact peak $1 - N^{-1/(N-1)}$.

47 Page 4: frontier amplification, water-filling, and scope remarks

Proposition 2’s hard-tail ratios are useful. Correct the mastered-tail RLOO statement. Proposition 3 should be a one-step mass allocation theorem under fixed pass rates, not a universal curriculum optimum.

Remark 1 is directionally right but can become shorter after the factorization theorem: mass controls estimator activity; transfer and sequential value remain outside the scalar. Remark 2 correctly admits that an adaptive curriculum changes the training mixture; keep this and consider making it more prominent.

The claim that a true-pass-rate oracle ties the posterior should be rewritten. The reported oracle ties posterior plus recycling, while oracle-only materially exceeds posterior-only.

48 Page 5: method diagram and hindsight proposition

The method diagram should show K' after relabeling. A dead requested-task group cannot simply be declared an all-success destination group under centered MaxRL, because that would also produce zero coefficients.

Replace Proposition 4 with theorem 15.1. State whether the destination is selected from the same group and whether the proof assumes cross-fitting or a fixed destination.

49 Page 6: partition map and recycling decomposition

“Creation is the only live channel” should be scoped to the practical estimator and the compared samplers. Add full-CV to the control battery.

The placebo decomposition is excellent. Keep the exact numbers together: replay captures most of the fixed-pool AUC gain, while the true relabel direction adds a smaller reliable increment. Do not lead with the cosine table after finding null relabels with the same cosine.

50 Pages 7–9: maze results and run counts

The central maze claim should be the estimator coverage main effect. The statement “whether the teacher amplifies GRPO’s decay is a factorial question” is already appropriately cautious; align the abstract and introduction with that caution.

Explain exactly why five MaxRL runs enter the nine-run complete-separation test and eighteen enter the level-resolved analysis. Identify seeds, warmstarts, and configurations. Use cluster-aware inference.

The wall-clock decomposition is useful. Elevate the three-currency analysis to a figure: wall-clock, steps, and generated tokens/transitions.

51 Page 10: GSM8K

Keep the endpoint sign and steering-dose observation, but frame the run primarily as a treatment-delivery failure. The teacher visits too few prompts to become decisive. Add uniform-with-replacement and stronger steering before using this as a headline interaction.

The pass@k harness discrepancy must be reconciled. Until then, quote only within-harness contrasts and label them secondary.

52 Pages 11–12: Countdown sharpening and gate

The sharpening result is one of the paper’s strongest new empirical observations. Lead with the replicated three-seed mean-up/coverage-down trade.

The mechanism sentence “a self-achieved value is at $p \approx 1$ ” is too strong. Show the destination-pass-rate distribution. Use destination-marginal shift and saturation as hypotheses supported by telemetry.

Separate tier 1 and the frontier tier. Report the decay bug as an unintended operating point and rerun the corrected sweep.

53 Page 13: limitations and Jugs boundary

The limitations section is unusually good. Strengthen it by separating possible causes in Jugs:

- narrow template support;

- low within-group trajectory diversity;
- shared-parameter interference;
- capacity or representation limits;
- destination-key collision in the gate.

Do not let “the band collapses” absorb all of these mechanisms into one phrase.

54 Pages 14–16: appendices, hyperparameters, and reproducibility

The classic-control result is useful as a boundary demonstration that sparse verifier-style reward can create a strict zero-update freeze and that shared task conditioning is necessary. It is not central enough to carry the main narrative unless the LLM evidence is removed.

The hyperparameter table should distinguish derived quantities from tuned defaults. “Gate max $p = 0.5$, derived zero at $p \rightarrow 1$, midpoint tuned” is accurately described as motivated plus tuned, not derived.

Add a theory-to-code contract table and a run manifest. The statement that every figure is artifact-backed is strong; make the mapping from figure panel to run IDs visible.

Part VII

Final reviewer-style verdict and checklist

55 Reviewer-style verdict

55.1 Strengths

- A coherent estimator-first explanation of finite-group curriculum activity.
- A useful implementation-level refinement of practical MaxRL when scoped to the correct estimator.
- An exact closed-form coefficient profile and a stronger factorization connecting it to the finite- N gradient weight.
- Strong negative-result discipline, placebo controls, and correction history.
- A compelling empirical phenomenon: relabeling can improve mean accuracy while reducing multi-sample coverage.
- Good mechanistic instincts: requested-only posterior updates, conditioning rewrite, dose controls, and step-matched analysis.
- Boundary suites that make the applicability conditions visible rather than hiding failures.

55.2 Current weaknesses

- The central quantity is named more strongly than the theorem supports.
- The dead-zone impossibility omits the baseline’s full control-variate alternative.
- The curriculum-by-estimator interaction is not yet conclusively identified.
- Hindsight exactness remains too broad relative to source/destination distribution shift.
- The primary gate evidence uses an unintended effective setting.
- Statistical independence and run inclusion are not transparent enough.
- Preliminary LLM and inference results receive too much headline weight.
- The manuscript is not anonymous in its present artifact form.

55.3 Bottom line

Main takeaway

This is a major-revision paper, not a conceptual restart. The best version stops trying to prove that one scalar decides every aspect of curriculum learning and instead establishes a precise decomposition: the estimator defines finite-group activity; task structure defines transfer; relabeling changes the trained destination distribution; and coverage reveals when either channel concentrates rather than expands behavior.

56 Final pre-submission checklist

- ☐ The PDF and artifact are genuinely anonymous.
- ☐ Raw, full-CV, and practical MaxRL estimators are distinguished everywhere.
- ☐ Tajwar et al.’s raw $T = N$ theorem is not described as incorrect.
- ☐ A_N denotes full mass and $\nu_N = A_N/2$ denotes normalized activity.
- ☐ “Learning signal” is replaced by coefficient mass/activity, with the factorization theorem added.
- ☐ The dead zone is budget-indexed and scoped to the practical estimator.
- ☐ The full-control-variate all-fail baseline is included or its omission is explicitly justified.
- ☐ The exact peak is used; “band width” is defined or removed.
- ☐ The mastered-tail comparison with RLOO is corrected.
- ☐ Finite-group GRPO and population GRPO are clearly separated.
- ☐ Water-filling is described as a fixed-pass-rate one-step mass oracle.
- ☐ Oracle-only is compared correctly with posterior-only; oracle-plus-recycling is not used to dismiss information value.

- ☐ Estimator activity is separated from cross-task transfer.
- ☐ Hindsight is described as a destination semi-gradient under a proposal law.
- ☐ The relabeled success count K' and contrast mechanism are explicit.
- ☐ Same-group destination selection is discussed or cross-fitted.
- ☐ The gate is described as motivated by saturation and tuned in strength.
- ☐ Corrected multi-seed gate runs replace the bugged setting as the primary evidence.
- ☐ Maze statistics use seed/warmstart clusters and a public run registry.
- ☐ Five-versus-eighteen MaxRL inclusion rules are explained.
- ☐ “Six seeds” is changed to six contrasts within three seeds.
- ☐ Main effect and interaction claims are separated.
- ☐ Frozen-schedule estimator comparison is included.
- ☐ Uniform-with-replacement controls prompt repetition in GSM8K.
- ☐ Delivered steering is quantified by distributional metrics, not only dead-group fractions.
- ☐ Wall-clock, optimizer steps, and generated tokens/transitions are all reported.
- ☐ The pass@k evaluation harness discrepancy is reconciled.
- ☐ “Diversity premium” is renamed to a coverage-based term.
- ☐ Countdown figures separate saturated and frontier tiers.
- ☐ Destination-pass-rate and occupancy telemetry support the sharpening mechanism.
- ☐ The main paper contains at most three decisive experiment stories plus one boundary synthesis.
- ☐ Every load-bearing number appears in the paper or appendix, not only external Markdown files.
- ☐ Every theorem states its sampling law, estimator convention, and implementation assumptions.

A Compact derivation reference

For quick use while rewriting:

$$\begin{aligned}
p &= \mathbb{P}(r = 1 \mid x), & q &= 1 - p, & K &\sim \text{Bin}(N, p), \\
\mu_+ &= \frac{\nabla_{\theta} p}{p}, & \mu_- &= -\frac{\nabla_{\theta} p}{q}, & \mu_+ - \mu_- &= \frac{\nabla_{\theta} p}{pq}, \\
\mathbb{E}[\hat{g}_N^{\text{raw}}] &= \frac{1 - q^N}{p} \nabla_{\theta} p = \sum_{j=0}^{N-1} q^j \nabla_{\theta} p, \\
\mathbb{E}[\hat{g}_N^{\text{full}}] &= \mathbb{E}[\hat{g}_N^{\text{raw}}], \\
\mathbb{E} \sum |w_i|_{\text{full}} &= 2q - q^N, \\
\nu_N^{\text{prac}}(p) &= q - q^N = \text{pass}@N(p) - p, \\
\mathbb{E} \sum |w_i|_{\text{prac}} &= 2(q - q^N), \\
\mathbb{E}[\hat{g}_N^{\text{prac}}] &= \frac{1 - q^{N-1}}{p} \nabla_{\theta} p = \sum_{j=0}^{N-2} q^j \nabla_{\theta} p, \\
p_N^* &= 1 - N^{-1/(N-1)}, \\
\nu_N^{\text{RLOO}}(p) &= pq, \\
\nu_N^{\text{GRPO}}(p) &= \frac{1}{N} \mathbb{E} \sqrt{K(N - K)}, \\
\ell_N(p) &= 1 - p^N - q^N, \\
\nu_{n+1}(p) - \nu_n(p) &= pq^n.
\end{aligned}$$

B Suggested notation table

Symbol	Meaning
p	requested-task pass rate under the current policy
$q = 1 - p$	failure probability
N	rollout group size
K	observed number of successes in a group
S_i	trajectory score $\nabla_{\theta} \log m_{\theta}(z_i \mid x)$
$A_N(p)$	expected full coefficient mass $\mathbb{E} \sum_i w_i $
$\nu_N(p)$	normalized half-mass $A_N/2$; sampling activity utility
$\ell_N(p)$	probability of a nonconstant group $0 < K < N$
μ_+, μ_-	success- and failure-conditioned score means
$Q^{x \rightarrow g}$	source-induced proposal law after relabeling to destination g
Π^g	fresh destination policy law
K'	success count after destination rewrite and re-verification
ρ	fixed evaluation/task distribution
q_t	adaptive curriculum sampling distribution at step t

C Source map

The review relies on these locations in the supplied documents:

- Working draft page 1: abstract, title, public repository link, and headline claims.

- Working draft pages 2–4: research questions, contribution list, $T = N - 1$ lemma, coefficient mass, estimator comparisons, water-filling, and scope remarks.
- Working draft page 5: FrontierMax algorithm and hindsight proposition.
- Working draft pages 6–12: chain/frontier/maze/GSM8K/Countdown evidence and the gate bug disclosure.
- Working draft page 13: limitations, Jugs boundary, and correction history.
- MaxRL baseline page 5: raw success-conditioned estimator and Theorem 2, order $T = N$.
- MaxRL baseline page 6: full control variate and practical Algorithm 1 that drops $K = 0$.
- MaxRL baseline page 7: population weight-function view.
- MaxRL baseline pages 30–32: formal proofs and MaxRL weight derivation.

D References

References

- [1] Anonymous authors. *The Estimator Decides: What Curricula and Failure Recycling Can and Cannot Do in RL with Verifiable Rewards*. Working draft v0.9, August 4, 2026.
- [2] Fahim Tajwar, Guanning Zeng, Yueer Zhou, Yuda Song, Daman Arora, Yiding Jiang, Jeff Schneider, Ruslan Salakhutdinov, Haiwen Feng, and Andrea Zanette. *Maximum Likelihood Reinforcement Learning*. arXiv:2602.02710v1, 2026.